Course Project

Introduction

Background

As I was reading through several cases, this particular article immediately piqued my interest. It was refreshing to find a case study for public health and community outreach, especially after working on the last few case study homeworks. I am reminded that analytics is merely a tool, and as someone who is personally invested in how analytics can enact positive social change, I am excited by its versatility.

This project is centered around an SAS Analytics customer success story, "From data to dignity: How advanced analytics can help lift people from poverty," in which data analytics was used to help a non-profit organization make data-driven decisions and tailor its outreach programs and services.

Issue

Bissell Centre is a Canadian non-profit that services people in poverty, offering various lines of support spanning from basic necessities (food assistance, clothing, baby formula, etc.) to employment, housing, financial assistance, mental health, child care, and beyond. To support their diverse operations, Bissell Centre sought out a data analytic framework to evaluate the impact of their programs and services to advocate to policymakers for those in need and secure funding.

Challenges: Trust & Legality

Given that the nonprofit was working with a vulnerable population and presenting results to investors and policymakers, it was essential that the analysis avoided "black box" models and instead favored simpler, highly explainable and interpretable models for trust and transparency. Explainability may even be legally required.

Solution

The purpose of this report is to propose a framework of analytic models, required data and collection strategies, and refresh cycles that SAS likely used to support the following goals:

1. Identify individuals with need
2. Prioritize clients for intervention
3. Assign intervention to groups

Phase One: Classification Model

Purpose

Phase One of the analysis involves first identifying individuals with need. We can do this by predicting whether a person is insufficient in the key domains Bissell Centre supports (housing, employment, financial well-being, support network, mental health, physical health).

Data

In the article, Bissell Centre notes their intake assessment as one of their main methods of data collection. This assessment, also called the Self-Sufficiency Matrix, measures an individual's self-sufficiency in six domains: housing, employment, financial well-being, support network, mental health and physical health. Based on a a self-sufficiency matrix developed by Massachusetts' Department of Mental Health, rows in the matrix using ordinal data could look like the following:

Domain	1	2	3	4	5
Housing	Homeless or threatened with eviction.	In transitional, temporary, or substandard housing; and/or current rent/mortgage payment is unaffordable (over 30% of total income).	In stable housing that is safe but only marginally adequate.	Household is in safe, adequate subsidized housing.	Household is safe, adequate unsubsidized housing.
Employment	No job.	Temporary, part-time or seasonal; inadequate pay, no benefits.	Employed full time; inadequate pay; few or no benefits.	Employed full time with adequate pay and benefits.	Maintains permanent employment with adequate income and benefits.
Food	No food or means to prepare it. Relies to a significant degree on other sources of free or low-cost food.	Household receives SNAP.	Can meet basic food needs, but requires occasional assistance.	Can meet basic food needs without assistance.	Can choose to purchase any food household desires.

Using this structure, we can collect data about various aspects of each individuals' lives such as their income, childcare access, children's education, adult education, healthcare coverage, life skills, family/social relations, mobility/access to transportation, community involvement, parenting skills, legal/criminal background, mental health, physical health and disability, substance use, and domestic safety. Given that this matrix involves private health data, a

clinician or licensed professional may need to collect the data, and data privacy constraints such as HIPPA may need to be considered throughout the process.

Additionally, we can also collect data on the individual's demographics such as age, gender, ethnicity, marital status, etc. and their historical data (prior housing loss and crises, past self-sufficiency matrix scores) and previous programs/services usage (shelter stays, employment supports used, program participation).

Recommendation

Logistic Regression Classification

With historical, binary-outcome-labeled client data from the Self-Sufficiency Matrix or previous caseworker or administrative records, we can use a logistic regression classification to predict whether or not a new client is insufficient in a given domain to obtain assistance.

Threshold Selection/Optimization

Cost Considerations

Misclassification errors and cost need to be considered when determining a threshold. For this riskier decision, false negatives (i.e. failure to identify someone in need) are far more detrimental and costly than false positives (i.e. allocate resources to someone who may not need them). Therefore, the threshold should be set such that even individuals not at risk are misclassified, adopting the less conservative approach.

Optimization Approach

We can utilize the cost associated with false negatives and false positives in an optimization model to find the optimal threshold in which the expected cost is minimized, subject to constraints such as policy priorities, resource, and capacity constraints (the max number of people the non-profit can service a month, the number of caseworkers available, max capacity per program, etc.).

Alternatively, we could also use k-fold cross-validation to tune the threshold, so long as we have enough historical data to split it into different sets (training, validation, and testing).

Evaluation of Model

Advantages

With a logistic regression classification model, we can evaluate the model quality on test data with pseudo R-squared value, ROC and AUC, and a confusion matrix. Because of its involvement in high-stake decisions, logistic regression is a suitable model as it is simple, effectively handles binary outcomes of self-sufficiency, transparent, easy to interpret, and easy to explain to policyholders why an individual was classified as needing support.

Disadvantages

Although using ROC and AUC as estimates of model quality is quick, it cannot differentiate between the cost of false negatives and false positives, thereby, an inefficient metric when used alone. Therefore, evaluation of confusion matrix output for specific situations (e.g. sensitivity, specificity, etc.) is also proposed to help determine which model will provide the highest value for the organization.

Extension

Although this analysis aims to prioritize transparency for legality and working with policymakers/investors, we can supplement the logistic regression model by comparing the result with that of decision trees or random forests. Although these models are not as easily interpretable or explainable (random forest especially being a “black box” model), they may capture more nuanced patterns and produce a more robust, proactive solution in addition to the logistic regression model alone.

Phase 2: Survival Analysis

Purpose

Whereas Phase One focused on predicting who is at risk and at what probability, Phase Two focuses on prioritizing clients for intervention. Although we could use the results of the logistic regression and service clients in order of risk probability, it may be more beneficial to prioritize clients based on the imminence of their crises and predict when they are likely to experience a crisis.

Data

In addition to demographics; socioeconomic indicators; historical, time-stamped crises data (e.g. past instances of homelessness, shelter stays, job loss, evictions, etc.); and the risk probabilities from the logistic regression output, we can also utilize historical clients’ follow-up assessments at the three-, six-, and nine-month intervals mentioned in the article that captures program-specific outcome information.

Recommendation

Cox Proportional Hazards Model

Given the data collected above, we can use a Cox Proportional Hazards model to estimate the time until an individual may experience housing loss, job loss, relapse, or other crisis events and identify which cases are the most urgent to address.

Evaluation of Model

Advantages

Cox Proportional Hazards models are widely used in social science applications for examining the impact of socioeconomic factors on survival times, making it ideal for this case study. Furthermore, this model is equipped to handle censored data, which may be common for this population (e.g. clients leave the program early, lose contact and stop reporting, or do not experience any crises during data collection). Hazard ratios are also easily interpretable, aligning with our transparency goals for decision-making and reporting to policymakers and investors.

Disadvantages

In order to accurately calculate time-to-crisis, this model relies on clearly-defined and well-documented historical data. This requirement makes collecting data for this model particularly difficult, however, survival models such as Cox Proportional Hazards can handle incomplete outcome data so long as its data includes timestamps (e.g. intake date, program enrollment data, last follow-up date, program completion date, etc.).

Phase Three: Clustering

Purpose

After identifying individuals with need and ranking clients by risk and urgency, the last phase of the analysis focuses on assigning clients to the appropriate intervention programs or services.

Data

We can collect client demographics, historical data, risk probabilities from the logistic regression classification, and time-to-crisis estimates and hazard ratios as predictors clustering groups of clients by similar risk profiles and needs.

Additionally, the article mentions that during the intake process and at follow-ups, interviews are conducted and case notes are collected. These qualitative data sources could be extremely valuable if we used natural language processing techniques to transform the text into embeddings or topic features and extract themes and signals of risk that may otherwise have gone uncaptured by numeric data in phase one and two. By using the extracted features, we can enhance our clustering model and improve its accuracy.

Recommendation

Given the data discussed above, we can use a clustering model such as K-Means or a graph model with Louvain algorithm to group clients by similar risk and needs and help match them to a program.

Evaluation

Advantages

Because it is an unsupervised learning model, using a clustering model on the data for program assignment could also potentially discover new groups. This would help the nonprofit identify blind spots in their program offerings and allow them to renovate their services to assist these new groups.

Disadvantages

On the other hand, because it utilizes unsupervised learning, human interpretation and judgement of the clusters to programs is necessary.

Refresh and Re-Run Strategy

In general, as the non-profit collects more data on their clients, the models will need to be re-trained and the data refreshed to learn from recent experiences and improve its performance.

For each model in this analysis, it is likely that they may need to be re-run regularly at either monthly or quarterly intervals depending on how many new clients the non-profit takes on, the duration of the follow-up cycle, the volume of new case notes and interviews, the non-profits' needs, the cost of re-running/re-training models, and how dynamic the client demographic or policy is.

One approach to finding the optimal re-run frequency for each model would be to perform a simulation in which we adjust the frequency of re-running the model until the performance begins to degrade. This frequency rate would be the point at which our model starts to get outdated and balances costs with profits.

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